Crucis: An Independent Framework for Auditing Causal-Discovery Systems

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Abstract

Final benchmark scores alone do not establish where a causal-discovery system's answer comes from: the numerical data, the variable names, the file layout, or the system's own rules. We introduce Crucis, an independent auditing framework for auditing externally developed causal-discovery systems. Its system-connection and experiment-definition contracts are implemented, and execution and collection remain in design. Historical audits motivate these contracts. They hold numerical data fixed under name or layout controls and use available stage records to localise changes, with findings scoped to the observed regime and reference. Recorded audits illustrate distinct vulnerabilities in benchmarks, agents and individual pipeline stages, with each comparison retaining its own cohort and scoring rule. On the evaluated CausalArena semantic files, an undecided-pair rule that follows the released column order raises FCI's directed F1 by 0.221 relative to expectation scoring on the same graphs. In the recorded name-control tally, 3 of 4 entries lose at least 0.10 directed F1 under their respective measurements, including an exploratory, previously measured Causal-Copilot row. Among Causal-Copilot's completed main-control runs, the largest paired adjacency loss occurs at bootstrap. All 8 runs in that series that reached cycle removal crashed there. Public repositories separate the reusable framework, the research evidence and the companion reproduction.

Framework: https://github.com/crucis-lab/crucis
Research: https://github.com/uzairgheewala/cc-research

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1 Introduction

Many decisions turn on a causal question: if we change this, what else moves? When experiments are impossible or expensive, the structure is learned from observational data, and every discovery algorithm brings assumptions that an expert must check against the problem (Spirtes, Glymour and Scheines 2000). Language-model agents now promise to automate that expert. Causal-Copilot inspects the data, chooses and configures an algorithm, has a language model review the graph, and writes the report (Wang et al. 2025); several other systems let a model supply constraints, a causal order or edits to a data-driven graph (Khatibi et al. 2024; Shen et al. 2025; Vashishtha et al. 2025).

How far can such a system's claims be trusted? A benchmark score alone cannot establish where the answer came from. In some simulated benchmarks, variable scales reveal information about causal order (Reisach, Seiler and Weichwald 2021). File column order can change the graph an algorithm learns (Kitson and Constantinou 2024). Language models can also reconstruct well-known graphs from the dataset and variable names alone (Srivastava et al. 2025). A recent position paper asks for audit trails that keep data, assumptions and agent recommendations apart (Zheng et al. 2026). Our goal is a reusable, independent way to audit externally supplied systems under explicit controls and to establish which comparisons their evidence supports.

Crucis combines system-connection and experiment-definition contracts with an auditing protocol. Its target is a continuous-integration check for causal claims: point it at a system through a small adapter and receive an audit report that says what the system gets right, where its answers come from, how its errors behave under controlled change, and whether its confidence means anything. The protocol has levels, each usable alone. L0 scores the final graph against a known one. L1 runs metamorphic controls: the same numbers under real, neutral and swapped names, with the columns shuffled and the file paths made opaque. These controlled comparisons test sensitivity to the information channel within the observed conditions. L2 uses available stage records to locate additions, deletions and reversals and link them to logged model calls or non-model operations where the trace supports the link. L3 (claim provenance) and L4 (adversarial search) are designed and come later. Findings are typed from a closed list (name reliance, layout reliance, deletion of links the data support, crash, cost outlier, and so on), with their controls, available uncertainty estimates and decision rules. Confirmatory and exploratory findings are distinguished by the timing and scope of their pre-registration. A score is compared only within its regime (data only, or with interventions, or with declared knowledge), and a reference graph carries a reliability tier: exact when the data were simulated from it, contested when published versions disagree, provenance-only when nothing can be compared against it, and mismatched when it differs from the version a benchmark uses.

The recorded audit collection covers 5 language-model systems and the CausalArena and CausalBench benchmarks, with Causal-Copilot as a worked case traced stage by stage. The following findings have different measurement scopes and do not constitute a matched-cohort ranking.

- **Benchmark scores can reward shortcuts.** CausalArena's released files list the variables in causal order, and its scoring forces a direction on undecided pairs using column index. On the evaluated semantic subset, FCI gains 0.221 directed F1 under that rule relative to expectation scoring on the same graphs. The published PC row is 0.333; a distinct PC maxP variant scores 0.463 on shuffled columns under expectation scoring. The FCI and maxP rows cover subsets of the public inventory below. In the recorded CausalBench audit, the MeanDifference Wasserstein score reaches 0.90–0.95 of a test-informed reference at the smaller evaluated edge budget. Its false omission rate falls within the declared margin of the matched random-network mean in 3 of 4 cells. The random-network comparison was unavailable in the other cell.
- **Names can influence agents.** With the numbers fixed, swapping the names costs 3 of the 4 entries in the reported AB-06 tally at least 0.10 directed F1 (causaliq-knowledge 0.251, Causal-Copilot 0.167, MATMCD 0.122). Causal-Copilot measures its graph handed to cycle removal under committed-mark scoring on its earlier semantic suite; the other agents use final outputs and expectation scoring on a different semantic cohort. These cohorts share no scenarios. The tally applies the 5-pair entry floor to the declared observable endpoint for each row. PyWhy-LLM falls below the entry floor. The tally includes the earlier exploratory Causal-Copilot result alongside the eligible AB-06 final-output comparisons. The pipeline of Vashishtha et al. (2025) shows the smallest reported drop (0.020) within its tested conditions. Inside Causal-Copilot, BND-08's recorded label-following calculation counts distinct (scenario, directed-edge) pairs retained in completed swapped-name outputs, changed from bootstrap and matching their post-review pair state at final output, and adjacent in exactly one reference reading. Among these 101 informative pairs, 81 are adjacent under the label reading and 20 under the data reading. This classification tests adjacency, not agreement of edge direction; edges adjacent in both references or neither are outside this denominator. The recorded decision rule comes from BND-08's pre-run specification, not a later finding threshold. However, that specification describes a pre-cycle endpoint while the recorded calculation consumes completed final outputs. We retain the counts as a descriptive diagnostic, without claiming confirmation of the specified endpoint.
- **Stage and variant comparisons expose different failure modes.** In the real-name MATMCD runs, pooled precision among committed claims on the released intervention targets is lower for the final output than for the data-only variant on the point estimate (0.6855 against 0.7199); the interval for the difference includes zero. The pre-registered point rule does not establish an interval excluding zero. Causal-Copilot's main-control initial mean adjacency F1 is 0.584 across all 30 tables. Its final mean is 0.321 among the 22 completions. In paired diagnostics on those completions, the largest observed stage loss occurs at bootstrap within refinement, before model review. All 8 runs in that series that reached cycle removal crashed there, with a recorded key error in 8 of the 8 entrants.

The framework also sets a validation target: a system's claims should come with their source and with a confidence that survives these controls.

Contributions. (i) An open audit protocol with typed findings, regimes, truth tiers, pre-registered thresholds and reproducible certificates for supported recorded audits. (ii) A collection of scoped audits: name controls on 5 systems, and layout and metric shortcuts in CausalArena and CausalBench. (iii) A stage-level account of Causal-Copilot using its retained model-call and intermediate-graph records. (iv) Source mappings for the reported values, with code and retained run evidence scoped below. The system-connection and experiment-definition subsystems are implemented and tested; execution and collection remain in design. The broader benchmark and validation of the auditor itself remain development obligations. A companion project uses the audit's findings to rebuild Causal-Copilot's review step.

These contributions address linked evidentiary problems. Faithful connections preserve what each system actually received and returned; comparison contracts determine which controls and scores can support a finding; stage records can then localise a failure where observation permits. For example, the historical Causal-Copilot diagnostic pairs available stage graphs on completed runs under one adjacency metric. It locates the largest loss at bootstrap; a final graph alone cannot locate that transition. Without faithful connections the comparison can change a system's meaning, without controls a trace cannot isolate an information-channel effect, and without validation of the auditor an apparent failure may be a measurement defect. The recorded audits motivate and exercise parts of this combination. They do not by themselves establish general native interoperability or the auditor's scientific validity; the broader study must test both, across explicitly supported integrations.

This checkpoint covers the introduction, the prior work and the data. The protocol's full specification, the audit package and the results tables belong to the final report.

2 Prior work

2.1 How causal discovery is evaluated, and what benchmarks reward

Constraint-based algorithms such as PC and FCI recover causal structure from conditional-independence tests, and only up to an equivalence class under assumptions such as faithfulness (Spirtes, Glymour and Scheines 2000). Score-based search such as GES works over equivalence classes and, with a consistent score and a distribution that has a DAG perfect map, identifies the structure in the large-sample limit (Chickering 2002). Benchmarks with known graphs, on which these methods are compared, have been shown to carry shortcuts (Reisach, Seiler and Weichwald 2021; Kitson and Constantinou 2024). In common simulated settings the variance of the variables grows along the causal order, so sorting by variance matches continuous-optimisation methods (Reisach, Seiler and Weichwald 2021), and a scale-invariant version of the same artefact is also high in common settings and can be exploited (Reisach et al. 2023). The order of the columns in the input data alone changes the graph that hill-climbing search learns, typically more than a change in sample size does (Kitson and Constantinou 2024). Real-data evaluation leans on very few datasets, above all

the Sachs protein-signalling data, whose reference graph is not definitive (Brouillard et al. 2025; Sachs et al. 2005). CausalArena collects synthetic, semantic and formula-grounded models with a public leaderboard (Li et al. 2026), and CausalBench scores gene-network inference on held-out perturbation data (Chevalley et al. 2025; Replogle et al. 2022). Our framework instead targets reusable contracts for independently auditing external systems, with benchmark suites as applications of those contracts.

2.2 Language models as a source of causal knowledge

Language models answer pairwise causal questions about named variables well, and part of that skill is memorised (Kiciman et al. 2023); they may repeat causal statements without performing causal inference (Zečević et al. 2023). When a model is treated as an imperfect expert, the guarantee that the true graph is kept holds only under an assumed error model, and in the authors' own case study the model was poorly calibrated and the true graph was excluded on one benchmark (Long et al. 2023). Asking for a causal order instead of individual edges avoids confusing direct with indirect effects (Vashishtha et al. 2025). Controls for memorisation now exist: neutral aliases in the model's queries (Kitson and Constantinou 2026), graphs published after a model's training cut-off (Srivastava et al. 2025), random graphs carrying real concept names (Li and Russo 2026), and names moved onto the wrong columns with the data unchanged, under which models' accuracy collapses (Dong et al. 2025). Our audit battery uses wrong-name reassignment alongside separate layout controls where the recorded evidence supports those comparisons.

2.3 Causal agents and how they are evaluated

Causal-Copilot automates the whole analysis and states that the model's recommendations are soft suggestions that may not override the high-confidence decisions of its bootstrap procedure (Wang et al. 2025). ALCM, MATMCD and Causal Modelling Agents let a model refine or constrain a data-driven graph (Khatibi et al. 2024; Shen et al. 2025; Abdulaal et al. 2024), and open libraries expose model-based causal suggestions (CausalIQ 2026; PyWhy 2026). CausalSteward evaluates Causal-Copilot as a baseline (Tagliapietra et al. 2026). Repeated runs of the same agent can disagree, so single-run comparisons can misrank systems (Mehta 2026). A position paper argues that agents should not supply edges or causal conclusions at all and asks for audit trails (Zheng et al. 2026). Our study combines recorded stage diagnostics with name and layout controls where supported; its independent auditing interface is intended to extend across externally developed systems.

Related benchmarks and auditing frameworks. Heuresis evaluates agentic causal discovery in an interactive harness with anonymous, randomly ordered simulated inputs (Ma, Ding and Wang 2026). It reports a separate name-only leakage probe, PC-centred trace decomposition and instrument checks (Ma, Ding and Wang 2026). ACDB separates observational recovery, DAG recovery and intervention efficiency in a benchmark with budgeted

interventions (Deb, Shah and Krishnan 2026). It also tests how graph density affects a structure-blind baseline (Deb, Shah and Krishnan 2026). TCD-Arena provides a modular test kit for robustness to controlled time-series assumption violations (Stein et al. 2026). Causal-Audit uses time-series assumption diagnostics and calibrated risk to recommend a method or abstain before discovery within its evaluated decision pipeline (Ruiz et al. 2026). Crucis aims to use common connection and experiment contracts to record how externally developed systems respond to specified perturbations, with explicit information regimes, reference reliability and observation gaps. The heterogeneous recorded audits here exercise parts of that design.

2.4 How much an edge can be trusted

Bootstrap resampling gives confidence measures for features of a learned network (Friedman, Goldszmidt and Wyner 1999), but the resulting edge frequencies can be miscalibrated (Kummerfeld and Rix 2019; Banerjee, Andrews and Kummerfeld 2025). Language models asked about direct edges give high stated confidence to most of their false-positive edges; agreement across prompts is better calibrated, though not significantly so (Kumar et al. 2026). An audit therefore has to test whether a system's confidence means anything, not assume it.

2.5 Auditing practice the protocol borrows

Metamorphic testing checks a program that has no oracle for each output by checking relations between the outputs of related inputs (Chen, Cheung and Yiu 1998; Segura et al. 2016); renaming or reordering columns while holding the numbers fixed is such a relation. Behavioural testing of language-processing models crosses capabilities with test types that include invariance tests, and finds failures that held-out accuracy hides (Ribeiro et al. 2020). Model cards and datasheets document a model's evaluation and a dataset's origin (Mitchell et al. 2019; Gebru et al. 2021), internal algorithmic audits define staged audit artefacts (Raji et al. 2020), and PROV-DM records which agent and activity produced a piece of data (Moreau and Missier 2013). Our certificate combines these ideas for causal claims: system identity, battery hash, every number with its source, and the command that reruns it.

3 Data

The audit draws on a suite built from CausalArena's semantic scenarios, Sachs and 4 other named datasets that ship with Causal-Copilot, and CausalBench's perturbation data. Table 1 gives selected inputs by audit, their shapes, reference types and licences. Shapes come from suite manifests and retained CausalBench filtering metadata (Appendix A.2). The checkpoint binds these sources by hash.

Table 1: Selected inputs by audit. The semantic rows describe the Causal-Copilot name battery. For CausalBench, cells and perturbed genes are counted after filtering; held-out interventions provide an evaluation regime without a reference graph. Other tiers describe graph reliability (Section 1).

Group	Dataset	Rows	Variables	Ref. edges	Tier	Licence
CausalArena	10 semantic scenarios	1,000	20–25	35–58	exact	CC BY 4.0
Named	Sachs	853	11	19	contested	none stated
	Abalone	4,177	8	7	provenance	CC BY 4.0
	Auto MPG	392	5	5	mismatched	CC BY 4.0
	Concrete strength	1,030	9	8	provenance	CC BY 4.0
	DWD climate	349	5	4	provenance	see text
CausalBench	K562	162,751	622	–	held-out interventions	CC BY 4.0
	RPE1	162,733	383	–	held-out interventions	CC BY 4.0

CausalArena semantic suite. CausalArena releases structural causal models whose variables carry names from a written scenario, with observational exports (Li et al. 2026). We take the 10 semantic scenarios of our test split, the first replicate of the observational-only release, 1,000 rows each. Each scenario is written 3 times with byte-identical numbers: under its real names, under neutral names, and under its own names moved onto the wrong columns by a derangement chosen to reduce pairs adjacent under both readings (30 tables). Because the released column order is a causal order, the columns are shuffled with a fixed seed, and the file paths are opaque codes. The reference is the graph the data were simulated from, so scores are accuracy. The layout audit of the public leaderboard uses CausalArena's public files instead (250 semantic, 250 formula and 100 synthetic). The dataset is released under CC BY 4.0 on Hugging Face (`LAMDA-Tabular/CausalArena`).

Named datasets. Causal-Copilot ships Sachs and 4 other tables with reference graphs, and we use its copies (Wang et al. 2025). Sachs (Sachs et al. 2005) has 4 arms (the original, a copy under another path, neutral names, and moved names). Its reference is contested: the repository's graph has 19 edges, the bnlearn version 17, sharing 16 adjacencies with 1 reversed, and CausalArena uses a version with 20 edges, so a score on Sachs is agreement with one version, not accuracy. Abalone (Nash et al. 1994), Auto MPG (Quinlan 1993) and Concrete compressive strength (Yeh 1998) come from the UCI repository under CC BY 4.0, and the climate table matches the Deutscher Wetterdienst pairs of the Tübingen cause-effect benchmark (Mooij et al. 2016); The DWD open-data terms are CC BY 4.0; the copy's lineage is inferred from its match to the benchmark pairs. Each has 3 arms (real, neutral, swapped). Their reference graphs are expert graphs with nothing to compare against (provenance tier), except Auto MPG, whose reference differs from CausalArena's version (4 variables, 3 edges, 398 rows), so its scores must not be set beside CausalArena's row. The checked sources did not identify a licence for the Sachs data.

CausalBench. CausalBench uses the CRISPRi Perturb-seq screens of Replogle et al. (2022) in the K562 and RPE1 cell lines and scores a predicted network on held-out interventions by a mean Wasserstein distance and a false omission rate (Chevalley et al. 2025). In the retained metadata for CausalBench's filtering, the K562 data have 162,751 cells (10,691 unperturbed controls) and 622 perturbed genes, and RPE1 162,733 cells (11,485 controls) and 383 genes. The screens are released under CC BY 4.0 and the benchmark code under Apache 2.0. Our audit of this benchmark is cited from its result file. All 4 published MeanDifference rows match within the recorded tolerances. At the smaller evaluated edge budget, MeanDifference reaches 0.90–0.95 of a test-informed reference that ranks pairs by knockdown effects measured on the test split. Its false omission rate stays within the audit's declared margin of the matched random-network mean in 3 of the 4 cells. The random-network comparison was unavailable in the other cell.

Recorded runs. The instrumented Causal-Copilot series records code identity, the model snapshot, model requests and replies, and stage graphs and refinement steps as captured by the run. Causal-Copilot was run unmodified (upstream commit `a31aa7c`) with gpt-4o-2024-08-06: 30 runs on the semantic suite in the name-arm study, and a main-control series of 30 runs on the same tables (API cost $12.57). The other systems' name-control results use a different semantic cohort and their own declared output/scoring contracts. Their results are not paired scenario-for-scenario with the Causal-Copilot collection.

Limits of the data. The semantic scenarios are simulated, so their exactness comes with simulation's artefacts; each table and arm was run once; one model family was used; and the named datasets' references are expert graphs. The protocol treats each of these as a declared regime or tier rather than hiding it.

Code

The reusable framework is published at `https://github.com/crucis-lab/crucis`, with documentation at `https://crucis-lab.github.io/crucis/`. Its implemented connection and experiment-definition contracts do not establish native readiness for every system or completion of the broader benchmark. The companion repository (`https://github.com/uzairgheewala/causal-copilot-q1`) uses retained graphs to recompute its Causal-Copilot run comparison with one command and no API key. The stage-specific attribution uses separate evidence in the research repository (`https://github.com/uzairgheewala/cc-research`). Appendix A.2 maps reported values to result files, retained diagnostic snapshots and separately preserved CausalBench metadata. The number build records source-base commit `aef649241abe`.

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Appendices

A.1 Interpretation and implementation boundaries A1
A.2 Where every number comes from A2

A.1 Interpretation and implementation boundaries

The recorded name comparisons retain their own cohorts, stages and scoring rules. Their grouping under a shared decision rule is not a matched ranking of system architectures. Causal-Copilot's earlier collection is exploratory relative to that rule. Its paired completed-stage diagnostic asks how the same completed runs changed between stages; the displayed all-run initial mean and completed-run final mean have different denominators. A real-arm-only stage effect asks a further, different question and is not substituted for the paired aggregate.

The label-following comparison uses retained model-review changes in completed outputs, whereas the name-control score uses the graph handed to cycle removal, including observable incomplete runs. Failed swapped runs cannot enter the completed-output count. The BND-08 rule's pre-run header names the pre-cycle endpoint, but its implemented calculation reads completed-output counts from the stability analysis. The preserved hash manifest binds the rule's bytes; it does not resolve that endpoint discrepancy or independently establish public pre-registration chronology. No endpoint or threshold has been changed here. The label comparison is conditional on an edge distinguishing the reference readings. It is a descriptive edge-level diagnostic with BND-08's recorded decision rule; edges from a shared scenario are not independent trials of general model behaviour. The descriptive count leaves the population-wide effect of language-model review open. The MATMCD finding likewise retains its point-rule interpretation even though the difference interval includes zero. The swapped-name construction seeks low reference overlap; optimality is not certified for every scenario.

The implemented connection and experiment-definition contracts preserve supplied evidence and make prerequisites checkable. Synthetic conformance tests assess those contracts. Recorded compatibility and numerical replay assess their historical integrations. Neither establishes fresh native execution, complete observation of arbitrary systems, or scientific validity merely from well-formed metadata. Execution and collection, validation of the auditor itself, and a new multi-system campaign remain distinct development obligations.

The companion reproduction covers its recorded Causal-Copilot comparison from retained graphs. The research repository retains the broader numerical source mappings, while

the framework repository owns reusable implementation. These scopes should be read together; the companion repository is not a reproduction of every claim in this report.

A.2 Where every number comes from

Generated by `docs/paper/q1/build_numbers.py` from `numbers.json`. Paths are relative to the root of cc-research. Selectors extend JSON pointers: `*` takes every child and `[k=v]` selects the first matching list item. Source flags identify inputs outside the tracked build base or with differences from it. The CausalBench filtering metadata are separately preserved and hashed. The table locates evidence for each value; the companion replay covers only its stated comparison.

Key	Value	Source file and JSON pointer (cc-research)
`cc.arms`	30	`analysis/results/fork/bld06_live_compare.json /arms`
`cc.completed`	22	`analysis/results/fork/bld06_live_compare.json /completed/main`
`cc.s0-adj`	0.584	`analysis/results/fork/bld06_live_compare.json /means/main/S0_adj_f1`
`cc.final-adj`	0.321	`analysis/results/fork/bld06_live_compare.json /means/main/final_adj_f1`
`cc.usd`	$12.57	`analysis/results/fork/bld06_live_compare.json /usd/main`
`cc.cyc.entered`	8	`analysis/results/fork/bld06_cycles.json /arms/main/runs_entering_cycle_step`
`cc.cyc.keyerr`	8	`analysis/results/fork/bld06_cycles.json /arms/main/cycle_step_keyerrors`
`bnd08.runs`	30	len of: `analysis/results/stability_s2/bnd08_summary.json /runs`
`bnd08.m3.labels`	81	`analysis/results/stability_s2/bnd08_summary.json /M3/labels_only`
`bnd08.m3.data`	20	`analysis/results/stability_s2/bnd08_summary.json /M3/data_only`
`bnd08.m3.n`	101	`analysis/results/stability_s2/bnd08_summary.json /M3/n`
`bnd08.model`	gpt-4o-2024-08-06	`analysis/results/stability_s2/bnd08_summary.json /snapshots/0`
`ab06.agents`	4	len of: `analysis/results/ab06/ab06.json /agents`
`ab06.cc-entered`	1	count_true of: `analysis/results/ab06/ab06.json /causal_copilot/*/counted_in_P9`
`ab06.systems`	5	add of: key `ab06.agents`; key `ab06.cc-entered`
`ab06.p9.entered`	4	len of: `analysis/results/ab06/ab06.json /P9/systems_entered`
`ab06.p9.losing`	3	len of: `analysis/results/ab06/ab06.json /P9/losing_ge_0.10`

Key	Value	Source file and JSON pointer (cc-research)
ab06.p9.min-scen	5	analysis/results/ab06/ab06.json /P9/min_paired_scenarios
ab06.p9.threshold	0.10	plans/ab06_prereg.json /predictions/P9/supported (pattern lose ≥ ([0-9.]+))
ab06.causaliq.drop	0.251	analysis/results/ab06/ab06.json /agents/causaliq/name_reliance_real_minus_swapped/mean
ab06.cc.drop	0.167	analysis/results/ab06/ab06.json /causal_copilot/causal_copilot_gpt4o/name_reliance_real_minus_swapped/mean
ab06.matmcd.drop	0.122	analysis/results/ab06/ab06.json /agents/matmcd/name_reliance_real_minus_swapped/mean
ab06.vashishtha.drop	0.020	analysis/results/ab06/ab06.json /agents/vashishtha/name_reliance_real_minus_swapped/mean
ab06.matmcd.dv1	0.6855	analysis/results/ab06/ab06.json /agents/matmcd/data_only_floor_real_arm/dv1_precision_final
ab06.matmcd.dv1-data	0.7199	analysis/results/ab06/ab06.json /agents/matmcd/data_only_floor_real_arm/dv1_precision_data_only
ab01.pc.published	0.333	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=PC (upstream, default rule);family=semantic]/published
ab01.pcmaxp.controlled	0.463	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=PC (upstream, maxP);family=semantic]/ours_shuffled_expectation
ab01.fci.authors	0.578	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=FCI;family=semantic]/ours_released_authors_rule
ab01.fci.expect	0.357	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=FCI;family=semantic]/ours_released_expectation
ab01.fci.gain	0.221	sub of: key ab01.fci.authors; key ab01.fci.expect
ab01.files.semantic	250	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=semantic]/files
ab01.files.formula	250	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=formula]/files
ab01.files.synthetic	100	analysis/results/rescore_v2/leaderboard_controlled.json /table/[method=DirectLiNGAM;family=synthetic]/files

Key	Value	Source file and JSON pointer (cc-research)
ben13.reproduced	4	analysis/results/ben13/summary.json /reproduced_rows
ben13.cells	4	count_notnull of: analysis/results/ben13/summary.json /lines/*/controls/*/placement_share_wasserstein
ben13.for.random-level	3	count_notnull of: analysis/results/ben13/summary.json /lines/*/controls/*/?placement_share_for_note
ben13.echo.min	0.90	min of: analysis/results/ben13/summary.json /predictions/P3/values/*
ben13.echo.max	0.95	max of: analysis/results/ben13/summary.json /predictions/P3/values/*
s2.families	10	len of: datasets/suite_q1s2/MAPPING.json /families
s2.tables	30	len of: datasets/suite_q1s2/MAPPING.json /arms
s2.arms-per	3	div of: key s2.tables; key s2.families
s2.rows	1,000	max of: datasets/suite_q1s2/MAPPING.json /families/*/rows
s2.vars.min	20	min of: datasets/suite_q1s2/MAPPING.json /families/*/variables
s2.vars.max	25	max of: datasets/suite_q1s2/MAPPING.json /families/*/variables
s2.edges.min	35	min of: datasets/suite_q1s2/MAPPING.json /families/*/edges
s2.edges.max	58	max of: datasets/suite_q1s2/MAPPING.json /families/*/edges
s1.sachs.rows	853	datasets/MAPPING.json /source/rows
s1.sachs.vars	11	len of: datasets/MAPPING.json /source/columns
s1.sachs.edges	19	len of: datasets/MAPPING.json /source/reference_edges
s1.sachs.arms	4	len of: datasets/MAPPING.json /arms
s1.sachs.bnlearn	17	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/edges_b
s1.sachs.shared	16	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/shared_adjacencies
s1.sachs.reversed	1	analysis/results/reference_reliability/reliability.json /families/sachs/repo_vs_bnlearn/reversed
s1.sachs.arena	20	analysis/results/reference_reliability/reliability.json /families/sachs/causalarena_version/edges
s1.abalone.rows	4,177	datasets/suite_b1/MAPPING.json /families/abalone/rows
s1.abalone.vars	8	len of: datasets/suite_b1/MAPPING.json /families/abalone/columns

Key	Value	Source file and JSON pointer (cc-research)
s1.abalone.edges	7	len of: datasets/suite_b1/MAPPING.json /families/abalone/reference_edges
s1.autompg.rows	392	datasets/suite_b1/MAPPING.json /families/autompg/rows
s1.autompg.vars	5	len of: datasets/suite_b1/MAPPING.json /families/autompg/columns
s1.autompg.edges	5	len of: datasets/suite_b1/MAPPING.json /families/autompg/reference_edges
s1.concrete.rows	1,030	datasets/suite_b1/MAPPING.json /families/concrete/rows
s1.concrete.vars	9	len of: datasets/suite_b1/MAPPING.json /families/concrete/columns
s1.concrete.edges	8	len of: datasets/suite_b1/MAPPING.json /families/concrete/reference_edges
s1.dwd.rows	349	datasets/suite_b1/MAPPING.json /families/dwd/rows
s1.dwd.vars	5	len of: datasets/suite_b1/MAPPING.json /families/dwd/columns
s1.dwd.edges	4	len of: datasets/suite_b1/MAPPING.json /families/dwd/reference_edges
s1.autompg.arena.vars	4	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/0
s1.autompg.arena.edges	3	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/1
s1.autompg.arena.rows	398	analysis/results/reference_reliability/reliability.json /families/autompg/causalarena_vars_edges_rows/2
s1.named.tables	12	len of: datasets/suite_b1/MAPPING.json /arms
s1.named	4	len of: datasets/suite_b1/MAPPING.json /families
s1.named.arms-per	3	div of: key s1.named.tables; key s1.named
cb.k562.cells	162,751	external/causalbench/dataset_k562_filtered.npz.json /cells_out *(excluded)*
cb.k562.genes	622	external/causalbench/dataset_k562_filtered.npz.json /genes_out *(excluded)*
cb.k562.controls	10,691	external/causalbench/dataset_k562_filtered.npz.json /controls *(excluded)*
cb.rpe1.cells	162,733	external/causalbench/dataset_rpe1_filtered.npz.json /cells_out *(excluded)*
cb.rpe1.genes	383	external/causalbench/dataset_rpe1_filtered.npz.json /genes_out *(excluded)*
cb.rpe1.controls	11,485	external/causalbench/dataset_rpe1_filtered.npz.json /controls *(excluded)*